

Lecture No. 2 :

=> Modules

=> Pip

Modules :

→ Basically used to borrow other's code.

→ We can use code which is written by another coder in our python code using Modules.

→ In Python, a **module** is a file that contains ready-made functions, classes, and code which we can use in our program.

Instead of writing everything from scratch, we can import modules and use their features directly.

→ Two types of Modules :

How modules help us

- Save time
- Reduce code length
- Make programming easier
- Reuse code again and again
- Provide powerful ready-made features

→ By which we can easily work on heavy projects and build them.

i) Build in Modules (install using pip)

ii) External Modules (install using pip)

External Modules : are modules made by other developers that are not built into python by default. We must install them before using them.

Example → pyttsx3 (this module converts text into speech)

install :

pip install pyttsx3

Program :

import pyttsx3

```
engine = pyttsx3.init()
```

```
engine.say("Hello Deepak")
```

```
engine.runAndWait()
```

Output :

Computer speaks : "Hello Deepak"

Other External Modules Examples :

numpy → Mathematics

pandas → Data analysis

pygame → Games

requests → API & web requests

flask → Web Development

Build in Module : in python are modules that already come with Python. We do not need to install them.

Example → math (used for mathematical operations)

```
import math
```

```
print(math.sqrt(25))
```

Output : 5.0

Other common build-in modules :

random → Generate random numbers

os → Work with files/folders

datetime → Date & Time

time → Delay and timing

statistics → Mean, Median and Mode

Another Example :

```
import random
```

```
print(random.randint(1,10))
```

this prints a random number between 1 and 10

pip is the **standard package manager for Python**. It is used to install, upgrade, and manage additional libraries and dependencies that are not included in the standard Python distribution. [[1](#), [2](#), [3](#), [4](#)]

Practical :

Install Pandas.

Commands to use in terminal in VS Code :

pip install pandas

A screenshot of a Windows command prompt window. The prompt is 'PS D:\Python>' and the command 'pip install pandas' is being entered. The text is displayed in a monospaced font with a color scheme where 'pip' is yellow and 'install pandas' is white on a black background.

```
PS D:\Python> pip install pandas
```

Install sklearn

pip install scikit-learn

Build in module, no need to install directly use it

import hashlib

Install tensorflow

pip install tensorflow
